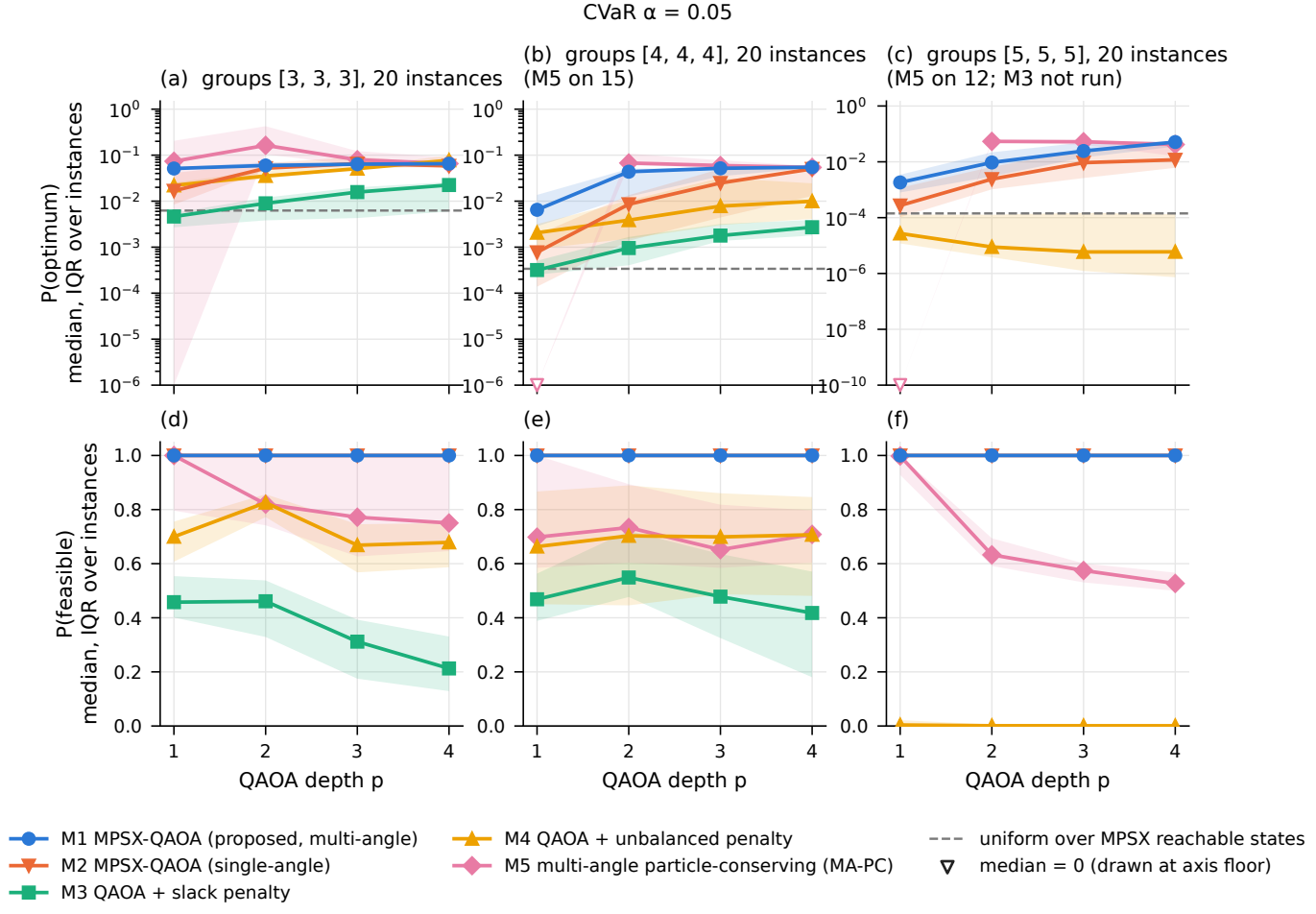
CVaR α = 0.05
(a) groups [3, 3, 3], 20 instances
(b) groups [4, 4, 4], 20 instances
(M5 on 15)
(c) groups [5, 5, 5], 20 instances
(M5 on 12; M3 not run)
P(optimum)
median, IQR over instances
(d)
(e)
(f)
P(feasible)
median, IQR over instances
QAOA depth p
QAOA depth p
QAOA depth p
M1 MPSX-QAOA (proposed, multi-angle)
M2 MPSX-QAOA (single-angle)
M3 QAOA + slack penalty
M4 QAOA + unbalanced penalty
M5 multi-angle particle-conserving (MA-PC)
uniform over MPSX reachable states
median = 0 (drawn at axis floor)